MIL-C-3659A
24 June 1968
SUPERSEDING
MIL-C-3659
8 January 1952

MILITARY SPECIFICATION

CRUCIBLE, FILTERING, LABORATORY, Gooch, Low Form, Fritted, 30 ml.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers 30-milliliter (ml.), low form, fine porosity, fritted, Gooch crucibles, suitable for the collection of fine precipitates by filtration and drying in laboratory procedures.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

DD-G-541	Glass Laboratory
MMM-A-250	Adhesive, Water-Resistant, for Sealing Fiberboard Boxes
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

FSC 6640

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 Marking for Shipment and Storage."

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) E-128;

Maximum Pore Diameter and Permeability of Rigid Porous Filters for Laboratory Use.

Applications for copies should be addressed to the

AMERICAN SOCIETY FOR TESTING AND MATERIALS
1916 Race Street, Philadelphia, Pennsylvania, 19103

3. REQUIREMENTS

3.1 Material. Material shall be borosilicate glass conforming to Type I, class A of DD-G-541.

3.2 Description

3.2.1 Construction. Crucible shall be all glass, low form, Gooch type with a fine porosity fritted disk fused to the body. Crucible shall have the following dimensional characteristics:

Top outside diameter (below bead)	40 ± 1 mm.
Bottom outside diameters (at point directly below the fritted disk)	
a. Top	29.5 ± 1 mm.
b. Bottom	25.5 ± 1 mm.

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Overall height	50 mm. nominal
Disk thickness	2 to 2 1/2 mm.
Disk diameter	30 mm. nominal
Wall thickness	1 1/2 to 2 mm.

3.2.2 Pore size. Pore size of fritted disk shall be between 3.5 and 5.5 microns when tested as specified in 4.4.1.

3.3 Finish. Top and bottom edges shall be fire-polished. All other surfaces (except that of the disk) shall be smooth.

3.4 Identification marking. Crucibles shall be permanently and legibly marked with the name or registered trade-mark of the manufacturer.

3.5 Workmanship. Workmanship shall be first class throughout. The crucibles shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

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	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

Component (Unit of product)	Characteristic	Requirement	Test Procedure	Inspection level	AQL
Crucible	Pore size	3.2.2	4.4.1	I	*

* Acceptance number shall be zero.

4.3 Examination. The crucibles shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of Defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major
- 101 Disk not uniform in structure.
- 102 Disk not free of chips (see 203).
- 103 Crucible not clean, dry, and ready for use.
- 104 Disks not free of black spots (see 201).

Minor
- 201 Disk not free of a black spot (see 104).
- 202 Surfaces, except disks, not smooth.
- 203 Crucible (other than disk) not free of chips (see 102).

* Examination shall not be restricted to the classified possible defects listed above.

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4.3.2 Dimensional examination. The crucible shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Pore size test. Pore size shall be tested, using water as the test liquid, in accordance with ASTM, E-128, Maximum Pore Diameter, and Permeability of Rigid Porous Filters for Laboratory Use.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One crucible, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	576 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be over-packed in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676 except that commercial colors are acceptable. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents Closure shall be adequate to prevent spilling of contents under normal handling.

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5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type I, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with appendix of applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Closure shall be in accordance with appendix of applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

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5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.2.1 and 5.2.2).

6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-3659A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426 REPLACES EDITION OF 1 OCT 54, WHICH MAY BE USED.

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